Finite normal pairs as incidence geometries

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Abstract

We give a complete reconstruction of the finite normal-pair representation results of Díaz Ramos, Molinier and Viruel. An explicit ordered-pair graph encodes a finite group and a normal subgroup, and the common chamber theorem turns it into a connected incidence geometry. We also recover the authors' smaller recursive (S_n, A_n) family, including its color sizes and faithful top action. A completed-hexagon example is calculated directly. Its corrected type-preserving subgroup, the chamber degree correction and the degree-one base of the recursive family are kept separate from the unchanged principal representation theorem.

1 An explicit representation of a normal pair

We use the finite colored-graph definition of incidence geometry and the groups $C(X), K(X)$ from the common paper [2]. Correlations permute color classes; $K(X)$ fixes them individually. For a finite group G and a normal subgroup H, the following is Theorem A of Díaz Ramos, Molinier and Viruel [1], with the rank and connectedness of their construction stated explicitly.

Theorem 1. *Every finite normal pair $H \trianglelefteq G$ is isomorphic to $(C(\Gamma), K(\Gamma))$ for a finite connected incidence geometry Γ. For $G \neq 1$ the construction below has rank $[G : H] + 3$; for $G = 1$ it has rank one. No fixed rank-two assertion is made.*

Normality is necessary because $K(\Gamma)$ is the kernel of the palette action. In particular G/H embeds in a symmetric group on the palette, so rank two would force $[G : H] \leq 2$. This elementary restriction is not evaded by allowing correlations that exchange types.

Theorem 2 (Ordered-pair encoding)**.** *Let $G \neq 1$ be finite and $H \trianglelefteq G$. There is a connected properly colored triangle-free graph $X(G, H)$ of minimum degree two with*

$$(C(X(G,H)), K(X(G,H))) \cong (G, H).$$

For $q = |G|$ it has $(q^3 + 9q^2 - 8q)/2$ vertices and $[G : H] + 3$ colors.

Proof. This is the explicit construction of [1, Theorem 3.1]. Enumerate $G = \{g_1 = 1, g_2, \dots, g_q\}$. Take a vertex P_x for each $x \in G$. Label the ordered pair (x, y), for $x \neq y$, by k when $yx^{-1} = g_k$. Thus $k \geq 2$. Introduce separate vertices

$$T_{x,y}, \quad S_{x,y,0}, \dots, S_{x,y,k+2}$$

for every ordered pair. Suppressing the pair subscripts within one gadget, add exactly the edges

$$P_x S_0, \quad S_0 P_y, \quad S_i S_{i+1}\ (0 \leq i < k+2), \quad P_x T, \quad T S_{k+2}, \quad S_{k+2} P_y. \tag{1}$$

There are no edges between original vertices and no other edges. Color P_x by its coset xH. Give all T-vertices one new color, the even-indexed S-vertices a second, and the odd-indexed ones a third. All three colors occur and are disjoint from the coset palette.

Every displayed edge joins different colors: only consecutive chain vertices use the two parity colors together, and every other edge has distinct vertex kinds or joins the old and new palettes. A

single gadget has no triangle. Its short paths between the old endpoints have lengths two and three, while the chain between S_0, S_{k+2} has length $k + 2 \geq 4$. More directly, no adjacent pair of internal neighbors of either old endpoint occurs in the edge list, and the internal chain with its one attached T has no triangle. Distinct gadgets meet only at old vertices, which are pairwise nonadjacent, so no triangle can use different gadgets. Every pair of old vertices is connected through its gadget and every new vertex connects to its endpoints. Thus the graph is connected.

The degrees of old vertices are $4(q - 1)$, those of S_0, S_{k+2} are three, and all remaining vertices have degree two. Consequently the old vertices are exactly those of degree at least four, including when $q = 2$. Let an uncolored graph automorphism induce $P_x \mapsto P_{\rho(x)}$. The degree-two neighbors of P_x are precisely the $T_{x,z}$. Among them $T_{x,y}$ is the unique one at distance two from P_y: its other neighbor is the terminal vertex $S_{x,y,k+2}$, adjacent to exactly that old vertex P_y. The terminal vertex is also the unique degree-three neighbor of $T_{x,y}$.

Starting at this terminal vertex, the path with degree-two internal vertices that ends at a degree-three vertex is intrinsic. It is the chain ending at $S_{x,y,0}$ and has length $k + 2$. The other route through $T_{x,y}$ ends at an old vertex of degree at least four. Hence the graph recovers both the orientation and the full label k of each ordered pair. Once the old permutation is fixed, all gadget vertices are forced. Conversely a permutation of the old vertices preserving these labels extends uniquely by preserving every gadget index. This is exactly the recognition and extension mechanism of the common theorem, applied first to the labeled ordered-pair structure.

Label preservation is the equation

$$\rho(y)\rho(x)^{-1} = yx^{-1} \qquad (x \neq y).$$

Taking $x = 1$ gives $\rho(y) = y\rho(1)$ for $y \neq 1$, and the same equality holds at $y = 1$. Thus the full uncolored automorphism group consists of right translations. For ordinary composition of permutations use the homomorphism

$$g : P_x \longmapsto P_{xg^{-1}},$$

with the same right translation on both indices of every gadget. The inverse in this formula fixes the left/right convention; it does not alter the graph construction.

All translations preserve the three auxiliary colors. Normality of H makes them permute the cosets xH, so every uncolored automorphism is a correlation. A translation fixes all coset colors if and only if $g \in H$: necessity follows from the coset of the identity, and normality gives sufficiency on every coset. This proves the exact pair assertion.

For each label $k = 2, \ldots, q$ there are q ordered pairs, with $k + 4$ internal vertices per pair. The total is

$$q + q \sum_{k=2}^{q} (k + 4) = \frac{q^3 + 9q^2 - 8q}{2}.$$

This is the size of the borrowed construction, not an optimality statement. □

Proof of Theorem 1. For $G \neq 1$, apply the common chamber-completion theorem to $X(G, H)$. It preserves the correlation pair, palette and connectedness, so the resulting geometry has rank $[G : H] + 3$. For $G = 1$, use a single vertex with one color. Its correlation group and type-preserving subgroup are both trivial, and its only nonempty clique is a chamber. □

The common cleaning theorem supplies a separate conclusion: any nonempty finite incidence system, including one with isolated vertices or triangles, can first be cleaned and then completed without changing its pair. Cleaning preserves components rather than making a disconnected input connected. Thus the connectedness asserted above comes from the explicit encoder; it is not an unstated hypothesis or conclusion of the arbitrary-input case.

2 The recursive symmetric-group family

The smaller family in [1, Example 5.2] contains explicit information not implied by the bare existence theorem. We retain its construction, color counts and faithful top action.

Theorem 3. *For each $n \geq 2$ there is a connected properly colored triangle-free graph X_n, with colors $0, \dots, n$, satisfying*

$$(C(X_n), K(X_n)) \cong (S_n, A_n).$$

Its color sizes are

$$|t^{-1}(n)| = n, \qquad |t^{-1}(i)| = \frac{n!}{2(i-1)!} \quad (0 \leq i < n), \tag{2}$$

where $(-1)! = 0! = 1$. The top class carries the faithful natural S_n-action. An odd permutation exchanges exactly colors zero and one, while an even permutation fixes every color. The minimum degree is at least two for $n \geq 3$.

Proof. Let X_2 be the four-vertex path with successive colors $2, 0, 1, 2$. Label its endpoints $1, 2$. Its only correlations are the identity and the reversal; reversal exchanges colors zero and one. This gives the claimed pair and faithful top action. The endpoints have degree one, so this is a pre-induction object, not a minimum-degree-two base case.

Suppose X_{n-1} has been constructed, with its top vertices labeled $1, \dots, n-1$. Introduce vertices $v_1, \dots, v_n$ of color n and vertices $v_{\{i,j\}}$ of color $n-1$, joined to v_i and v_j. Choose even permutations $g_j \in A_n$ with $g_j(n) = j$ and $g_n = 1$; these exist because A_n is transitive for $n \geq 3$. At each j attach a copy of X_{n-1}, identifying its top vertex labeled i with $v_{\{g_j(i),j\}}$. The interiors of different copies are disjoint, and all their lower colors are inherited.

The graph is connected: each copy is connected and meets its boundary, and the new top and pair vertices form the subdivided complete graph. Within every copy the coloring is proper and there is no triangle. Different copies meet only at pair vertices, all of the same color and hence pairwise nonadjacent. The neighbors of a new top vertex also all have color $n-1$ and are pairwise nonadjacent. Thus no triangle is introduced. An interior vertex retains its old degree. A top vertex has degree $n-1$; a pair vertex gets its two new top neighbors and its neighbors in two copies. For $n = 3$ only pair vertices had been endpoints of the base paths: their new degree is four, whereas the other vertices have degree two. For larger n the minimum-degree assertion follows by induction.

There are n top vertices and $\binom{n}{2}$ pair vertices. For $i < n-1$, the class of color i is the disjoint union of n copies of the previous class. These recurrences and the base path prove (2).

For $\sigma \in S_n$ set $v_i \mapsto v_{\sigma(i)}$ and $v_{\{i,j\}} \mapsto v_{\{\sigma(i),\sigma(j)\}}$. On the copy at j use the inductively constructed correlation corresponding to

$$h_{\sigma,j} = g_{\sigma(j)}^{-1} \sigma g_j.$$

This permutation fixes n and therefore acts on the $n-1$ labels of that copy. Its sign equals the sign of σ, because the chosen representatives are even. The induction hypothesis says that all these local maps have the same palette permutation: the identity for even σ, or exchange of zero and one for odd σ.

On a boundary vertex labeled i in the copy at j, the image has global label

$$\{g_{\sigma(j)} h_{\sigma,j}(i), \sigma(j)\} = \{\sigma g_j(i), \sigma(j)\}.$$

Thus local maps agree with the displayed map on every shared pair vertex. They combine into a correlation, with inverse given by σ^{-1}. Moreover

$$h_{\tau,\sigma(j)} h_{\sigma,j} = h_{\tau\sigma,j},$$

so their composition respects multiplication. We have constructed an S_n-action with the stated palette kernel and natural action on the top.

It remains to exclude extra correlations. For $n \geq 4$, the top class of size n is uniquely smallest, since the next size is $\binom{n}{2} > n$ and all lower classes are larger. For $n = 3$ all four classes have size three, but the top vertices are precisely the degree-two vertices both of whose neighbors have degree four. Each other degree-two vertex has one neighbor of degree two and one of degree four. Therefore every correlation preserves the top class, in every case.

A correlation fixing the top pointwise fixes each pair vertex, since its two top neighbors characterize it. We verify the component-recognition hypothesis needed before applying induction to a copy. The graph X_m with its top removed is connected for every $m \geq 2$. For $m = 2$ it is an edge. For $m \geq 3$, it is the union of its m connected attached copies, and any two copies meet in their common pair vertex. Hence it is connected.

In X_n, deleting both top and pair vertices consequently leaves exactly the connected interiors of the attached copies. The boundary of the interior at j is

$$\{v_{\{i,j\}} : i \neq j\}.$$

Every boundary vertex meets the interior: in the base it is an endpoint of a path, and at every subsequent stage each top vertex has neighbors in the lower part. These boundary stars are distinct for $n \geq 3$. A correlation fixing all pair vertices must therefore preserve each copy. On that copy it is a correlation fixing the top pointwise, and is the identity by inductive faithfulness. This is the component-boundary rigidity argument of the common paper.

Thus restriction embeds the full correlation group in S_n. The constructed action already realizes every member of S_n, and its palette kernel is A_n. This proves all claims. □

For $n \geq 3$, the common chamber theorem immediately produces (S_n, A_n)-geometries of rank $n+1$. For $n = 2$, first clean the base path and then complete it, or use Theorem 1. The source's blanket minimum-degree claim must be qualified at this base: it is not legitimate to apply the chamber theorem directly to its degree-one endpoints.

3 A completion that enlarges the group

Number the vertices of a hexagon by $i \in \mathbb{Z}/6\mathbb{Z}$, color them by parity, and add a third-colored tip on each edge $\{i, i+1\}$ with i even. This is [1, Example 5.1]. The three tips have degree two and the six hexagon vertices degree three, so every correlation recognizes the hexagon. Its allowed dihedral symmetries are exactly the stabilizer of the alternating edge set. They are generated by $i \mapsto i+2$ and $i \mapsto 1-i$, and form S_3. The first generator preserves the two old colors and the second exchanges them; the pair is (S_3, C_3).

Now add a third-colored tip on all six edges. Every edge lies in its displayed triangle, and those triangles are the maximal cliques; the completed system is an incidence geometry. Old vertices now have degree four and tips degree two, so again the hexagon is intrinsic. Every dihedral symmetry extends uniquely to its incident tips and permutes the parity classes. Consequently the full correlation group is the dihedral group of order twelve. The color-preserving subgroup consists of rotations by even increments and reflections $i \mapsto 2a - i$. It has order six and is isomorphic to S_3. In particular

$$i \longmapsto -i, \qquad \text{tip on } \{i, i+1\} \longmapsto \text{tip on } \{-i-1, -i\} \tag{3}$$

is a nonidentity type-preserving involution.

Thus the completed pair is (D_{12}, S_3), with the subscript denoting group order. The source prints (D_{12}, C_3). Formula (3) alone already excludes the second entry C_3; the intrinsic hexagon argument computes the full pair. The intended obstruction survives: naive completion of this system acquires additional correlations. The initial system already has triangles, so it is outside the common chamber theorem. This example does not prove that each of that theorem's hypotheses is necessary in every possible alternative construction.

4 Source contract and certificates

The source is Díaz Ramos–Molinier–Viruel, arXiv:2511.02410v2, 16 December 2025. The full finite normal-pair theorem, the ordered-arrow construction, chamber completion, cleaning and the recursive symmetric family belong to those authors. Their proof acknowledges Frucht and König's classical graph-representation constructions. Those historical representation theorems are not unproved premises of the explicit constructions here.

The common paper and this application recover source Theorems 3.1, 4.1, 4.2 and A. They also recover the complete intended statements of Examples 5.1 and 5.2 with the stated corrections. All local recognition, extension, triangle-free, minimum-degree, connectedness and faithful-action arguments are supplied. The three qualifications are the actual chamber degrees, the completed-hexagon type kernel, and the degree-one recursive base path. No optimality assertion about all completion hypotheses or all possible graph sizes is inferred. The family is asserted as (S_n, A_n); no universal identification of that pair with $(\mathrm{Aut}(A_n), \mathrm{Inn}(A_n))$ is needed or claimed. The source's Question 2.4 is answered by its own Theorem A; this recovery is not a newly solved conjecture.

The generic formal file for the common theorem proves the restriction and extension isomorphism for actual permutation subgroups and an embedded vertex set. It does not verify the gadgets that discharge its hypotheses. The unchanged `CompletedHexagonCounterexample.lean` gives four reported certificates: preservation of adjacency and color by (3), its order two, and the absence of a nontrivial involution in C_3. These are not a Lean computation of the full graph automorphism group or a formal proof for all finite normal pairs.

The two complete GAP programs in the appendix check the two hexagon groups, all 25 properly colored graphs on at most three labeled vertices under cleaning and completion, seven normal-pair examples, four completed pair examples, and the recursive family for $2 \le n \le 5$. They include exact restriction actions rather than comparing only group orders. They are bounded validation, since the general constructions and both exceptional group identifications have full written proofs above. The full programs remain available; no diagnostic code is removed. Their drivers and proof infrastructure are recorded separately. There is no numerical compression claim in this edition.

References

[1] A. Díaz Ramos, R. Molinier and A. Viruel, *Every finite group is represented by a finite incidence geometry*, arXiv:2511.02410v2, 16 December 2025.

[2] Human Compression Project, *Recognizable extensions of colored graphs*, accompanying common paper, Sections 1–3.

[3] R. Frucht, *Herstellung von Graphen mit vorgegebener abstrakter Gruppe*, Compositio Mathematica **6** (1939), 239–250.

[4] D. König, *Theorie der endlichen und unendlichen Graphen*, Akademische Verlagsgesellschaft, Leipzig, 1936.